

Student Number:

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2015

**YEAR 11**

HSC Assessment Point 1

# Mathematics

## General Instructions

- Date of task – Tuesday 17<sup>th</sup> November
- Weighting – 15%
- Reading time - 3 minutes
- Working time – 45 minutes
- Write using black or blue pen
- BOSTES approved calculators may be used
- A reference sheet is provided at the back of this paper
- Show all necessary working

## Total marks - 34

### Question 1-3

34 marks

- Start each question in a new booklet

### Outcomes

- Tangent to a Curve and Derivative of a Function
- The Quadratic Polynomial
- Locus and the Parabola



**Question 1 (10 Marks)**

- a) Find the equation(s) of a point  $P(x,y)$  that moves so that it is always 7 units from the line  $y = 2$ . 2
- b) Differentiate  $5x^3 - 2x + 4$  with respect to  $x$ . 1
- c) Find the equation of the tangent to the curve  $y = x^3 - 3x^2$  at the point  $(3, 0)$ . 3
- d) Find the equation of the directrix of the parabola  $x^2 = -18y$ . 2
- e) Find the values of  $k$  for which the quadratic equation  $2x^2 + 8x + k = 0$  has 2 distinct, real roots. 2

**Question 2 (14 Marks)      Start this question in a new booklet**

- a) Differentiate the following with respect to  $x$ , leaving your answer in fully factored form:
- (i)  $\frac{x-2}{x+3}$  2
- (ii)  $x^3(2x-9)^5$  3
- b) Let  $\alpha$  and  $\beta$  be the solutions of  $x^2 + 12x + 3 = 0$ . Find:
- (i)  $\alpha + \beta$  1
- (ii)  $\alpha\beta$  1
- (iii)  $\frac{1}{\alpha} + \frac{1}{\beta}$  2
- (iv)  $\alpha + \frac{3}{\alpha}$  2
- c) Find the values of  $a$ ,  $b$  and  $c$ , given that  $3x^2 - 14x - 2 \equiv a(x-2)^2 + b(x+4) + 1 + c$  3

**Question 3 (10 Marks)      Start this question in a new booklet**

- a)      Given that  $f(x) = \sqrt{4x+b}$  and that  $f'(11) = \frac{2}{5}$ ,
- (i)      Show that  $f'(x) = \frac{2}{\sqrt{4x+b}}$  2
- (ii)      Hence, find the value of  $b$ . 2
- b)      Consider the parabola  $y^2 - 4y - 6x - 8 = 0$ .
- (i)      Find the coordinates of the vertex 2
- (ii)      Find the coordinates of the focus 2
- (iii)      Show that the line  $3x + 4y - 3 = 0$  intersects the parabola exactly twice. 2

***End of Examination***

| Question 1 |                                                                                                                                                                 |                                                                                                                 |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Q          | Solution                                                                                                                                                        | Suggested marking criteria                                                                                      |
| A          | $y = 9$ and $y = -5$                                                                                                                                            | 1 mark for $y = 9$<br>1 mark for $y = -5$                                                                       |
| B          | $\frac{d}{dx}(5x^3 - 2x + 4) = 15x^2 - 2$                                                                                                                       | 1 mark for correct answer                                                                                       |
| C          | $y = x^3 - 3x^2$<br>$y' = 3x^2 - 6x$<br>When $x = 3$ ,<br>$y' = 9$<br>$y - 0 = 9(x - 3)$<br>$y = 9x - 27$                                                       | 1 mark for $y' = 3x^2 - 6x$<br><br>1 mark for $y' = 9$ at $x = 3$<br><br>1 mark for $y = 9x - 27$ or equivalent |
| D          | $4a = 18$<br>$a = \frac{18}{4}$<br>$= \frac{9}{2}$<br>Directrix equation is $y = \frac{9}{2}$                                                                   | 1 mark for the focal length, $a$<br><br>1 mark for the equation of the directrix                                |
| E          | Two distinct real roots<br>$\therefore \Delta > 0$<br>$\Delta = 8^2 - 4 \times 2 \times k$<br>$= 64 - 8k$<br>$\therefore 64 - 8k > 0$<br>$-8k > -64$<br>$k < 8$ | 1 mark for $64 - 8k > 0$<br><br>1 mark if answer follows                                                        |

| SECTION 2 |                                                                                                                                                                                                                                 |                                                                                                                                                    |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Q         | Solution                                                                                                                                                                                                                        | Suggested marking criteria                                                                                                                         |
| A i       | $\frac{d}{dx} \left( \frac{x-2}{x+3} \right) = \frac{1(x+3) - 1(x-2)}{(x+3)^2}$ $= \frac{5}{(x+3)^2}$                                                                                                                           | 1 mark for correct substitution in quotient or product rule<br><br>1 mark if answer follows                                                        |
| A ii      | $u = x^3 \quad v = (2x-9)^5$<br>$u' = 3x^2 \quad v' = 10(2x-9)^4$<br>$\frac{d}{dx} x^3 (2x-9)^5 = 3x^2 (2x-9)^5 + 10x^3 (2x-9)^4$ $= x^2 (2x-9)^4 [3(2x-9) + 10x]$ $= x^2 (2x-9)^4 (6x - 27 + 10x)$ $= x^2 (2x-9)^4 (16x - 27)$ | 1 mark for correct differentiation of $u$ and $v$<br><br>1 mark for correct substitution into product rule<br><br>1 mark for fully factored result |

|       |                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| B i   | $\alpha + \beta = -\frac{12}{1}$ $= -12$                                                                                                                                                                                                                                                                                              | 1 mark for correct answer                                                                                                      |
| B ii  | $\alpha\beta = \frac{3}{1}$ $= 3$                                                                                                                                                                                                                                                                                                     | 1 mark for correct answer                                                                                                      |
| B iii | $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$ $= \frac{-12}{3}$ $= -4$                                                                                                                                                                                                                                    | 1 mark for $\frac{\alpha + \beta}{\alpha\beta}$<br>1 mark for correct answer                                                   |
| B iv  | From ii<br>$\beta = \frac{3}{\alpha}$ $\therefore \alpha + \frac{3}{\alpha} = \alpha + \beta$ $= -12$                                                                                                                                                                                                                                 | 1 mark for $\beta = \frac{3}{\alpha}$<br>1 mark for correct answer                                                             |
| C     | $a(x-2)^2 + b(x+4) + 1 + c$ $= a(x^2 - 4x + 4) + b(x+4) + 1 + c$ $= ax^2 - 4ax + 4a + bx + 4b + 1 + c$ $= ax^2 + (b - 4a)x + (4a + 4b + 1 + c)$ $\therefore a = 3$ $\therefore b - 4a = -14$ $b - 12 = -14$ $b = -2$ $\therefore 4a + 4b + 1 + c = -2$ $12 - 8 + 1 + c = -2$ $5 + c = -2$ $c = -7$ $\therefore a = 3, b = -2, c = -7$ | 1 mark for correct expansion and factorising<br>1 mark for correct value of a and if b follows<br>1 mark if value of c follows |

| SECTION 3 |                                                                                                                                                                                     |                                                                                           |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Q         | Solution                                                                                                                                                                            | Suggested marking criteria                                                                |
| A i       | $f(x) = (4x + b)^{\frac{1}{2}}$ $f'(x) = \frac{1}{2}(4x + b)^{-\frac{1}{2}} \times 4$ $= 2(4x + b)^{-\frac{1}{2}}$ $= \frac{2}{(4x + b)^{\frac{1}{2}}}$ $= \frac{2}{\sqrt{4x + b}}$ | 1 mark for correctly showing use of the chain rule<br>1 mark for fully showing the result |

|       |                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                 |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| A ii  | $\frac{2}{\sqrt{4x+b}} = \frac{2}{5} \text{ when } x = 11$ $\frac{2}{\sqrt{44+b}} = \frac{2}{5}$ $\therefore \sqrt{44+b} = 5$ $44+b = 25$ $b = -19$                                                                                                                                                                      | <p>1 mark for <math>\frac{2}{\sqrt{44+b}} = \frac{2}{5}</math></p> <p>1 mark if solution follows</p>                                            |
| B i   | $y^2 - 4y = 6x + 8$ $(y-2)^2 = 6x + 12$ $(y-2)^2 = 6(x+2)$ <p>Vertex is at <math>(-2, 2)</math></p>                                                                                                                                                                                                                      | <p>1 mark for <math>y(y-4) = 6\left(x + \frac{4}{3}\right)</math></p> <p>1 mark if vertex follows</p>                                           |
| B ii  | $4a = 6$ $a = \frac{3}{2}$ <p>Focus is at <math>\left(-\frac{1}{2}, 2\right)</math></p>                                                                                                                                                                                                                                  | <p>1 mark for focal length following from b i</p> <p>1 mark if focus follows</p>                                                                |
| B iii | $3x + 4y - 3 = 0$ $3x = -4y + 3$ $6x = -8y + 6$ <p>Subs into parabola</p> $y^2 - 4y - (-8y + 6) - 8 = 0$ $y^2 - 4y + 8y - 6 - 8 = 0$ $y^2 + 4y - 14 = 0$ $\Delta = 4^2 - 4 \times 1 \times -14$ $= 16 + 56$ $= 72$ <p><math>\Delta &gt; 0 \therefore</math> two points of intersection between the line and parabola</p> | <p>1 mark for <math>y^2 - 4y - (-8y + 6) - 8 = 0</math> or equivalent</p> <p>1 mark for correctly using the discriminant to show the result</p> |